

Case Study: Warehouse Management System

WMS — a system used to track and manage the movement and storage of goods within a warehouse

Dan Whitfield has been running Whitfield Fulfillment for six years now, out of two rented warehouses: one in Reno, Nevada, and one in Columbus, Ohio. Small and mid-sized online sellers send their stock to whichever warehouse is closer, and Whitfield's team takes it from there.

Here's roughly how a day runs. When a shipment shows up, it usually comes with a tracking number from whichever carrier delivered it. If a seller just drops boxes off in person instead, someone writes down a ticket number for it. Whoever is on the receiving desk opens the boxes, scans the UPC barcode on each product with a handheld scanner, and logs how many came in, one row at a time, on that warehouse's Excel sheet. If something arrived crushed or damaged, it is supposed to get noted separately from the good stock, though in practice that just means another cell in the same row. Once everything from that shipment has been logged, the sheet gets marked "received," and the quantities are now supposedly available to sell.

On the way out it works in reverse. When one of the sellers' products sells, an order lands on a different tab. A staff member pulls the item, packs it, and weighs and measures the box so a shipping label can be printed. That step alone can take a day or two depending on how backed up things are. Once the label is on the box and it gets picked up, someone goes back and marks that row shipped.

For the most part this has worked, but lately it has started to fall apart. Two months ago a staff member's laptop froze while she was logging a large shipment, so she entered it again just to be safe. The sheet ended up showing twice the actual stock for three different products, and it took a manager most of a day to figure out why. Last month two different sellers' orders both got confirmed for the same nine units of a product that had already sold out, because two people happened to be editing the sheet on their own laptops at the same time, and neither one saw what the other had just changed. More than once Dan has asked "who changed this number" and gotten no real answer, because Excel just does not keep track of who typed what.

Dan is hiring more warehouse staff for both locations now, and he does not want everyone to have the same access to everything. The newer hires should not be able to see or touch what the two people he actually trusts can. A friend of his who works in tech mentioned a few things that might be worth adding, if they are possible. Staff keep calling him with the same kinds of questions, so maybe some kind of assistant could answer those instead. A manager should be able to find out how much stock they actually have without opening a spreadsheet. Staff scanning boxes usually have their hands full, so being able to talk to the system instead of typing into it would help too. His friend also mentioned that some warehouses let an AI handle simple tasks on its own now, and that there might be a way for Dan's one tech-savvy employee to script routine checks instead of doing them by hand every morning.

Dan does not know exactly what any of this should look like. He just knows Excel is not going to hold up much longer, and he wants a real system built for his two warehouses.